

- 111.** Sum of temperatures on 1st, 2nd, 3rd and 4th days
 $= (58 \times 4) = 232$ degrees ...*(i)*
 Sum of temperatures on 2nd, 3rd, 4th and 5th days
 $= (60 \times 4) = 240$ degrees ...*(ii)*
 Subtracting *(i)* from *(ii)*, we get:
 Temperature on 5th day – Temp. on 1st day = 8 degrees.
 Let the temperatures on 1st and 5th days be $7x$ and $8x$ degrees respectively.
 Then, $8x - 7x = 8$ or $x = 8$.
 $\therefore$ Temperature on the 5th day = $8x = 64$ degrees.
- 112.** Let A, B, C represent their respective weights. Then, we have:
 $A + B + C = (45 \times 3) = 135$...*(i)*
 $A + B = (40 \times 2) = 80$...*(ii)*
 $B + C = (43 \times 2) = 86$...*(iii)*
 Adding *(ii)* and *(iii)*, we get: $A + 2B + C = 166$...*(iv)*
 Subtracting *(i)* from *(iv)*, we get: $B = 31$.
 $\therefore$ B's weight = 31 kg.
- 113.** Let P, Q and R represent their respective monthly incomes. Then, we have
 $P + Q = (5050 \times 2) = 10100$...*(i)*
 $Q + R = (6250 \times 2) = 12500$...*(ii)*
 $P + R = (5200 \times 2) = 10400$...*(iii)*
 Adding *(i)*, *(ii)* and *(iii)*, we get $2(P + Q + R) = 33000$
 or $P + Q + R = 16500$...*(iv)*
 Subtracting *(ii)* from *(iv)*, we get $P = 4000$.
 $\therefore$ P's monthly income = ₹ 4000.
- 114.** Age of the instructor = $(34 \times 11 - 32 \times 10)$ years
 $= (374 - 320)$ years = 54 years.
- 115.** Monthly income of the married daughter
 $= ₹ (15130 \times 4 - 14660 \times 3)$
 $= ₹ (60520 - 43980) = ₹ 16540$.
- 116.** Weight of the teacher = $(35.4 \times 25 - 35 \times 24)$ kg = 45 kg.
- 117.** Age of the mother = $(12 \times 7 - 7 \times 6)$ years = 42 years.
- 118.** Total weight increased = (8×1.5) kg = 12 kg.
 Weight of the new man = $(65 + 12)$ kg = 77 kg.
- 119.** Total age decreased = (24×1) months = 24 months
 $= 2$ years.
 $\therefore$ Age of the newcomer = $(18 - 2)$ years = 16 years.
- 120.** Sum of the weights of the students after replacement
 $= [(52 \times 45) - (48 \times 5) + (54 \times 5)]$ kg = 2370 kg.
 $\therefore$ New average = $\left(\frac{2370}{45}\right)$ kg = $52\frac{2}{3}$ kg.
- 121.** Sum of heights of the 5 boys = $(25 \times 1.4 - 20 \times 1.55)$ m
 $= 4$ m.
 $\therefore$ Required average = $\left(\frac{4}{5}\right)$ m = 0.8 m.
- 122.** Total age decreased = (11×2) months = 22 months
 $= 1$ year 10 months.
 $\therefore$ Average age of two new players = $\left(\frac{35 \text{ years } 2 \text{ months}}{2}\right)$
 $= 17$ years 7 months.

- 123.** Let the original strength of the class be x .
 Sum of ages of the whole class = $(40x)$ years.
 Sum of ages of 12 new students = (12×32) years
 $= 384$ years.
 $\therefore \frac{40x + 384}{x + 12} = 36$
 $\Rightarrow 40x + 384 = 36x + 432$
 $\Rightarrow 4x = 48$
 $\Rightarrow x = 12$.
 Hence, the original strength of the class = 12.
- 124.** Let the original average weight of the class be x kg and let there be n students.
 Then, sum of weights of n students = (nx) kg.
 $\therefore \frac{nx + 50}{n + 1} = x + 1$
 $\Rightarrow nx + 50 = (n + 1)(x + 1)$
 $\Rightarrow nx + 50 = nx + x + n + 1$
 $\Rightarrow x + n = 49 \Rightarrow 2x + 2n = 98$...*(i)*
 And, $\frac{nx + 100}{n + 2} = x + 1.5$
 $\Rightarrow nx + 100 = (n + 2)(x + 1.5)$
 $\Rightarrow nx + 100 = nx + 1.5n + 2x + 3$
 $\Rightarrow 2x + 1.5n = 97$...*(ii)*
 Subtracting *(ii)* from *(i)*, we get: $0.5n = 1$ or $n = 2$.
 Putting $n = 2$ in *(i)*, we get: $x = 47$.
- 125.** Let there be n numbers in the original list and let their mean be x .
 Then, sum of n numbers = nx .
 $\therefore \frac{nx + 15}{n + 1} = x + 2$
 $\Rightarrow nx + 15 = (n + 1)(x + 2)$
 $\Rightarrow nx + 15 = nx + 2n + x + 2$
 $\Rightarrow 2n + x = 13$...*(i)*
 And, $\frac{nx + 16}{n + 2} = (x + 2) - 1$
 $\Rightarrow nx + 16 = (n + 2)(x + 1)$
 $\Rightarrow nx + 16 = nx + n + 2x + 2$
 $\Rightarrow n + 2x = 14$...*(ii)*
 Solving *(i)* and *(ii)*, we get: $n = 4, x = 5$.
- 126.** Let the average age of the whole team be x years.
 $\therefore 11x - (26 + 29) = 9(x - 1)$
 $\Leftrightarrow 11x - 9x = 46$
 $\Leftrightarrow 2x = 46 \Leftrightarrow x = 23$.
 So, the average age of the team is 23 years.
- 127.** Let the average for 20 innings be x . Then,
 $\frac{20x + 84}{21} = x + 2$
 $\Rightarrow 20x + 84 = 21x + 42$
 $\Rightarrow x = 42$.
 $\therefore$ Average after 21st inning = $42 + 2 = 44$.